Learning to Play

Noughts-and-crosses as an ecology of mortal scientists:
a worked example in which the generated logic pays its own way

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Abstract

A companion note [1] models learning as the loop the scientific method describes, run by a computation with a finite and conserved token supply in an environment made of other computations. Its worked ecosystem exhibits every construction of the framework as running code, and demonstrates none of the framework's reasoning: its hypotheses are arithmetic predicates on quanta, so nothing in it ever learns anything a reader would recognise as content.

This note supplies the missing demonstration by mapping a problem space whose correct answers the reader already holds. We take noughts-and-crosses, adversarial and solved, and build it as an energy landscape: an arena that holds no stack and merely routes, a stationary practice partner that is a source in the trophic sense, and a population of scientists whose hypotheses are formulae in the spatial–behavioural logic that the OSLF family generates from the presentation of the rho calculus [2]. Three things fall out of the mapping rather than being put in. A move is an assay, so playing and experimenting are one act. Minimax is an alternating modal formula, so the companion note's asymmetry between existential and universal modalities predicts that attack is learned before defence. And a fork—two threats at distinct squares—is a graded separating conjunction at $k = 0$, so the concept that game-playing programs normally hand-code is a term the generated logic already contains. That last demand dictates the board encoding: the squares, not the lines, must be the parands.

We then measure. A seven-rung ladder of hypotheses is priced by the modal depth of the formula each installs, and both the outcome distribution and the expected metering are computed exactly over the game's 255,168 plays. Three results are worth the price of the exercise. The spatial rung—the one a purely behavioural logic could not state—returns +0.0102 of win probability per metered unit, the best marginal return anywhere on the ladder and some five hundred times that of the modal rung below it. The exact characteristic formula χ is *dominated*: against fallible prey it wins less often than the heuristic rung beneath it while costing half again as much, which is the companion note's "yield, not truth" remark arriving as a number rather than a caveat. And a lineage whose theory is deep but whose opening is fixed cannot repair the opening by deepening; a crossover at the opening locus invades at identical cost, which is confinement, and its escape, measured on a real hypothesis space. We close on a prediction the conservation law forces: a population that solves its game stops eating. A solved science is a dead ecology.

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1 Introduction

1.1 Why a worked example, and why this one

The companion note [1] asks what it would take for a computation to do science, and answers in four parts: the environment of a computation is other computations; the hypothesis language is not chosen but generated from the presentation of the calculus; a hypothesis becomes an experiment by being installed as a guard, so that confirmation is a rendezvous; and the motive is metabolic, since tokens are conserved, never minted, and a learner that predicts badly starves.

Its Appendix A gives a small ecosystem in which every one of those constructions appears as ordinary code. What that appendix does not do—and the omission is the reason for this note—is show the framework *reasoning*. Its hypotheses are `q => q > 0` and `q => q > 4`: arithmetic predicates on the size of a quantum. No formula in it uses a structural connective, no assay discriminates between two structurally distinct hypotheses, the relaxation lattice is never walked with anything on it, and at no point does a better hypothesis visibly buy better performance. The appendix demonstrates that the framework *hosts* a learner. It does not demonstrate that it *produces* one.

The remedy is to instantiate the framework on a domain where the reader holds the correct answers independently, so that what the machinery emits can be scored against a standard the machinery did not supply. We take noughts-and-crosses. The choice needs a defence, since the game is trivial, and the defence is that triviality is the point: a worked example is for legibility, not difficulty, and a game small enough to solve exhaustively is a game in which every claim below can be checked rather than asserted. We checked all of them; §8 reports the numbers and Appendix B says how to regenerate them.

Remark 1 (The magic square comes free). The obvious alternative—a magic-square puzzle—is the same problem. In the game of *Fifteen*, two players alternately claim numbers from 1 to 9 and the first to hold three summing to fifteen wins. Writing the numbers into a Lo Shu square makes the triples summing to fifteen exactly the lines of a three-by-three grid, and the two games are isomorphic. We mention this because the puzzle framing is the one that would have been chosen for a solitary solver, and the isomorphism shows what that framing costs: it hides the opponent. An adversary is not an inconvenience here. It is the thing that makes the environment a computation with a strategy to be learned, which is the only kind of environment the companion note's ontology has.

1.2 What the mapping is supposed to show

Four features of the framework are exercised, and the interest is that each is *forced* by the domain rather than imposed on it.

A move is an assay. One cannot observe an opponent without playing, and the position afterwards is not the position before. §1.1 of [1] argues that observation is action; here it is not an argument but the rules of the game.

Minimax is an alternating modal formula. The modalities are labelled by redex positions, and §4.1 shows that in this domain a redex position *is* a move. So $\exists s.\langle K_s\rangle\forall t.[K_t]\cdots$ reads "I have a move such that for every reply I have a move such that…", which is minimax written out, and the quantifiers range over something rather than standing in for it. Proposition 6 of [1] then has something to say about which half of the game is cheap to learn.

A fork is a separating conjunction. Two threats whose completing squares are distinct. Disjointness is the entire content of the concept, and graded separation at $k = 0$ is exactly disjointness of interface. §4 shows that this demand dictates the board encoding.

Destructive assay bites. If winning harvests the loser, one never plays the same opponent twice, so hypotheses about individuals are worthless and the scientist must hold a hypothesis about a *namespace* of opponents.

1.3 What we are claiming, and what we are not

We claim that the framework, instantiated on this domain, generates a hypothesis language in which the recognised concepts of the game are terms rather than hand-coded features; that pricing those terms by modal depth produces a cost–benefit structure with a measurable and non-obvious shape; and that three of the companion note's propositions—the modal asymmetry, "yield, not truth", and confinement—are visible in the measurements rather than merely consistent with them.

We do not claim to have built a strong player. Noughts-and-crosses is solved and a lookup table plays it perfectly; nothing here competes with that and nothing should. Nor do we claim the pricing schedule is canonical: $\kappa(d) = 2^d$ is a choice, defended in §6.1 and varied in §8, and the qualitative results survive the variation while the exact boundaries do not. Finally, the guards below are written in the ideal logic. The current interpreter accepts only the arithmetic and pattern fragment of the `where` clause; where a guard uses a structural or modal connective it is a specification of a check rather than a check the reducer performs, and we mark each such occurrence.

2 What is imported

We recapitulate only enough to orient a reader who has not just read [1, 2], and refer to those notes for everything else.

The cut, and three grades of access. A world is $W \equiv S \mid E$: the scientist and its environment, separated by parallel composition. Computations are ontologically isolated, so E is other computations, and what S can come to know of E is bounded above by E's context-labelled bisimulation class. Access comes in exactly three grades: *perception*, the structural predicates evaluated at the surface of E, priced by a schedule $\kappa_{\text{sense}}(d)$ monotone in formula depth; *interaction*, the context-labelled modalities, costing a rendezvous per step; and *reflection*, the name predicate $@\varphi$ applied to names S already holds, which is cheap because it passes through no surface.

The logic is generated. Feeding the presentation of the `rho` calculus to the OSLF family yields one structural connective per term former, one context-labelled modality per rewrite rule and redex position, and—because $\mid$ carries an associative–commutative theory—a genuine separating conjunction:

$$P \models \varphi \mid \psi \iff P \equiv Q \mid R \text{ with } Q \models \varphi,\ R \models \psi.$$

Because a name is a quoted process, the name predicate $n \models @\varphi \iff n = @P$ with $P \models \varphi$ sees into the structure of a name; this is namespace logic [7]. The classifier note [2] adds two features we use: a greatest fixed point $\nu X.\varphi$, and *graded separation*

$$P \models \varphi \mid_k \psi \iff P \equiv \mathrm{H}\vec{a}.(Q \mid R), \quad Q \models \varphi,\ R \models \psi, \quad |\,\mathrm{fn}(Q) \cap \mathrm{fn}(R) \cap \vec{a}\,| = k,$$

which splits a term under restriction while counting the names the two halves share. Everything this note does that a modal logic cannot do is done with $|_k$.

Hypotheses are guards. The `where` clause of the rho calculus variant [4] puts a formula in operational position: $\texttt{for}(p \leftarrow c \texttt{ where } \vartheta)P$. A hypothesis installed as a guard is an experiment and a confirmation is a rendezvous; nothing is carried and no interpreter is needed, and the cost of testing is metered by the apparatus that meters everything else. An *assay* is a complementary guard pair, on φ and on $\neg\varphi$, whose three outcomes are $\mathsf{confirm}$, refute, and $\perp$—the last being the honest residue in which the environment did not answer in time.

The space and its metric. Beneath the characteristic formula χ_P sits a lattice of progressively weaker formulae obtained by forgetting conjuncts. Hypotheses live there, and revision is a move on it. The metric is ultrametric and stratified by the ν-unfolding stage: $d(\varphi, \psi) = 2^{-n}$ when the two agree on all approximants below stage n and differ at n. Since modal depth is what a finite budget bounds, distance, depth, and cost are one graded structure. The geometry is a tree, so there is no gradient, and Proposition 12 of [1]—*confinement*—says a walk whose steps are all shorter than 2^{-n} never leaves the ball of radius 2^{-n} it started in.

The game. Tokens are conserved: $\sigma + \kappa = \sigma_0 = \Theta$, with σ held in live stacks and κ migrated into the record. There is no primary producer. Under the internalising encoding $[\![-]\!] : \mathcal{C}(\mathsf{rho}) \to \mathsf{rho}$ a located stack becomes a message at a *metabolic channel* m_P, and *harvest* is an ordinary receipt on m_P performed by someone else. In $\mathcal{C}(\mathsf{rho})$ stacks are inviolable; in the image they are messages. The ecology lives in that gap, and Theorem 24 of [1] identifies the predatory contexts as exactly the non-image contexts holding a metabolic name. A *game*, in the technical sense of Definition 25 there, is a choice of admitted context class $\mathcal{A}$ with $[\![\mathcal{C}(\mathsf{rho})\text{-Ctx}]\!] \subseteq \mathcal{A} \subseteq \mathsf{rho}\text{-Ctx}$.

3 The world

3.1 Three kinds of thing, and only one of them holds tokens

The world contains an *arena*, a *practice wall*, and a population of *players*.

The arena implements the rules. It alternates moves, detects completed lines, and—this is the only thing about it that matters structurally—on a win it hands the winner the loser's metabolic name. It holds no stack of its own and it mints nothing. It *routes*.

That distinction is what keeps §16 of [1] intact. The objection to an exogenous reward signal is that it requires a referee holding a mint, and a referee with a mint destroys conservation and with it the entire motivational story. An arena that routes is not such a referee: every token it moves already existed, in the loser's stack, and the total is unchanged. What the arena supplies is not value but *permission*.

Observation 2 (The arena is the admitted context class). Definition 25 of [1] leaves $\mathcal{A}$ as a parameter and observes that its two endpoints are degenerate: at one end the decorated calculus with no game at all, at the other total war in which any context takes anything it can address. A game of noughts-and-crosses picks out a specific interior point. The contexts admitted to hold m_P are exactly those that have completed a line against P, and by Theorem 24 those are non-image contexts. So the rules of the game are not decoration on the ecology; the rules of the game *are* the choice of $\mathcal{A}$, and predation is legal here for precisely one reason, which is that somebody won.

Remark 3 (Capability gating, concretely). [1] distinguishes the capability-gated game—contexts that *derive* m_P from structure they can perceive—from one in which the name is simply handed over. Noughts-and-crosses is gated in exactly that sense once one asks how a player comes to complete a line: it must perceive the position, and perceiving the position is priced. The derivation is the play.

3.2 The wall is a source, not a prey

A world in which the only food is other players is a world of pure heterotrophs, and by §11 of [1] such worlds pay an unbounded epistemic rent and are hard to keep alive. We therefore include a *practice wall*: a stationary player that moves uniformly at random, backed by a finite reserve, which pays a small quantum per game played against it.

The wall has the access profile of a source in the precise sense of [1], §11.2: it is a persistent emitter, rate-limited by being a single contract, non-exclusive since anyone may call it, non-adaptive because its policy never revises, and exhaustible. A player is prey by the complementary profile: a linear send, exhaustible, exclusive, and adaptive. The distinction is not a difference of sort or of rewrite rule but of *multiplicity*, which is what makes the trophic split a fact about the cut rather than a taxonomy imposed on it.

This gives the world its two order parameters in interpretable form.

- δ, the concealment depth, is *how many plies of lookahead are needed to model one's opponent*. The wall has $\delta = 0$: its policy is uniform and there is nothing to learn. A player running the top rung has a δ of four or more.
- ι, the fraction of Θ already internal to scientists rather than loose in the environment, is the ratio of the players' stacks to the wall's remaining reserve. It rises monotonically as the wall is drained.

3.3 Conservation, stated for this world

Write $\Theta = R_0 + \sum_i \sigma_i(0)$ for the wall's opening reserve plus the players' opening stacks. Four operations move tokens and none creates them: the wall's payout moves reserve into a player; harvest moves a loser's stack into a winner's; endowment moves a parent's stack into a child's; and the metabolic debit moves a token from σ into κ. Hence $\sigma + \kappa = \Theta$ at every step, and the world is a finite race. The listing of Appendix A can be audited for this by inspection.

4 The board: the logic dictates the encoding

Here is the first place where the domain talks back, and we think it is the most instructive moment in the note.

We want "fork" to be expressible, because a fork is the first genuinely strategic concept in the game and because it is exactly the sort of thing a game-playing program hand-codes as a feature. A fork is a double threat: a position from which one has two distinct winning replies, so that the opponent can block only one. Disjointness is the whole content: two threats sharing their completing square are not a fork, they are one threat counted twice.

Separation is the connective for that. But which separation, and of what? The answer depends on how the board is encoded, and the constraint turns out to be tight.

The encoding. A board b carries, for each square k, two names built by reflection in the manner of the knot encoding's arcs: a *move channel* $sq_k = @[*b, k]$ and a *mark cell* $mk_k = @[*b, \texttt{"m"}, k]$. An empty square is a *receipt* offering to be claimed; an occupied square is the absence of that receipt together with a resting send on its mark cell:

$$\mathsf{Empty}_k \;\equiv\; \texttt{for}(@p \leftarrow sq_k \;\&\; @\_ \leftarrow mk_k)\{\; mk_k!(p)\;\} \qquad\qquad \mathsf{Held}_k(p) \;\equiv\; mk_k!(p).$$

In parallel sit eight *line* processes. A line reads the three mark cells of its line and replaces them; if it finds two marks of one player and one empty square s, it deposits a standing *offer* on a third family of names, $th_s = @[*b, \texttt{"t"}, s]$. A threat is therefore not computed on demand. It is a piece of the term.

Offers live on their own names rather than on the move channels for a reason worth stating: were a line to signal a threat by sending on sq_s, the signal would rendezvous with the square's receipt and the line would have played the move itself. Perception must not be able to act. That is the separation of grades one and two of Table 1 of [1], enforced by the choice of namespace.

Definition 4 (Threat, fork)**.** For a player p,

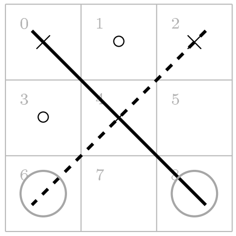
$$\mathsf{Threat}_s(p) := \mathsf{out}(th_s)@p, \qquad \mathsf{Threat}(p) := \mathrm{H}s.\,\mathsf{Threat}_s(p) \mid \top, \qquad \mathsf{Fork}(p) := \mathsf{Threat}(p) \mid_0 \mathsf{Threat}(p).$$

$\mathsf{Fork}(p)$ says the term splits into two parts, each carrying an offer, *sharing no restricted name.* Two threats at the same square—two lines with a common gap—both deposit on th_s, so the two parands share that name, $k \geq 1$, and $|_0$ excludes them. This is the graded separation of [2], §2.3.2, used at the bottom of its grading, and it was not introduced for this purpose.

Proposition 5 (The offers must be indexed by square)**.** *Let* $[\![B]\!]$ *be an encoding under which* $\mathsf{Fork}(p)$ *as given in Definition 4 is satisfied exactly by the positions in which* p *has a fork. Then the names over which* $|_0$ *counts sharing must be indexed by the completing square, not by the line.*

Proof sketch. Two lines of a three-by-three grid meet in at most one square, so distinct lines carry distinct line names and share none. Under a line-indexed encoding $|_0$ is therefore satisfied by any two distinct threats—including a pair whose gaps coincide, for instance the two diagonals when only the centre is empty. That is not a fork, since one move blocks both. Under the square-indexed encoding two threats share a name iff they share their completing square, so $k = 0$ holds iff the gaps are distinct, which is the definition of a fork. □

Remark 6 (What just happened). We did not choose the encoding and then ask what the logic could say about it. We asked for a concept, and the requirement that the generated logic express it determined the encoding. This is worth flagging because it is the mode in which the framework is meant to be used and it is the opposite of the usual order. In a hand-built game player, "fork" is a feature the programmer writes because they know the game; here it is a formula in a logic nobody designed, and the price of admission is a constraint on how the world is laid out. The companion note argues that a generated logic is the right hypothesis language on grounds of adequacy. This is the same claim cashed out as a design discipline.



threat A: line $\{0, 4, 8\}$, its only gap at square 8
threat B: line $\{2, 4, 6\}$, its only gap at square 6
the two lines *do* share square 4 — but it is occupied;
what they do not share is a *gap*, and $6 \neq 8$, so the two standing offers rest on different square channels:

$$\mathsf{Fork}(\times) \;=\; \mathsf{Threat}(\times) \mid_0 \mathsf{Threat}(\times)$$

Figure 1: A realised fork for crosses, and why $|_0$ at $k = 0$ is the right connective. Crosses threatens to complete on square 6 or square 8; noughts can block only one. What the separation must exclude is a pair of threats with a *common* gap, since one move blocks both—and that is exactly what $k = 0$ excludes once the squares, rather than the lines, are the parands (Proposition 5).

4.1 The redex context is the move

The modalities of the generated logic are labelled by contexts: one primitive $\langle K_j \rangle$ per rewrite rule *and choice of redex position*, where K_j is the one-hole context determined by that position ([2], §2.1). The label is not an action name. It records the surroundings in which the step fires, and in this domain those surroundings are exactly what carries the content.

The rho calculus has one interaction rule, so all the information in a label is in the position. Write a world in play as

$$W \;\equiv\; \mathrm{H}\vec{b}.\Big(\prod_{k \in \mathrm{empty}} \mathsf{Empty}_k \;\Big|\; \prod_{k \notin \mathrm{empty}} \mathsf{Held}_k(m_k) \;\Big|\; L \;\Big|\; Th \;\Big|\; M_p \Big),$$

with L the line processes, Th the standing offers, and M_p the mover's code, which for a move at s contains the send $sq_s!(p)$.

Definition 7 (The move context)**.** For an empty square s, the redex position of the claim at s is the join of $sq_s!(p)$ with Empty_s, and the one-hole context determined by it is

$$K_s \;:=\; \mathrm{H}\vec{b}.\Big([\,\cdot\,] \;\Big|\; \prod_{k \in \mathrm{empty},\; k \neq s} \mathsf{Empty}_k \;\Big|\; \prod_{k \notin \mathrm{empty}} \mathsf{Held}_k(m_k) \;\Big|\; L \;\Big|\; Th \;\Big|\; M'_p \Big),$$

M'_p being the mover's remaining code. We write $\langle K_s \rangle \varphi$ for the corresponding generated modality.

Proposition 8 (Moves are redex positions)**.** *The redex positions of W at which a claim can fire are in bijection with the empty squares, and the mover is determined by the context. Hence*

$$\text{“}p \textit{ has a move } s \textit{ after which } \varphi\text{”} \quad = \quad \exists s.\ \langle K_s \rangle \varphi,$$

where the existential ranges over redex positions, and the branching factor of that quantification is the number of legal moves.

Proof sketch. An occupied square offers no receipt, so no claim can fire there; each empty square offers exactly one, and the mark cells are disjoint, so the receipts do not race with each other. Distinct s give distinct contexts, since K_s retains $\mathsf{Empty}_{s'}$ for every $s' \neq s$ and $K_{s'}$ does not. Whose move it is, is fixed by which M_p sits in the context. □

Remark 9 (Why an action label would not do). Every claim, at every square, by either player, is the same action: a comm on a square channel carrying a mark. An action-labelled modal logic in the style of Hennessy–Milner sees one label and cannot distinguish nine moves; recovering the distinction would mean encoding the square into a name discipline and then reasoning about names, which is the manoeuvre [2], §3.1 declines in the knot setting for the same reason. Context-labelling gives it directly: the square being claimed is the hole, and everything else is the label.

Lemma 10 (The board settles)**.** *After a claim fires, the line processes run to a unique normal form in finitely many steps, and the resulting family of offers depends only on the marks.*

Proof sketch. Each of the eight lines performs one join on three mark cells and replaces what it took, so no line's output enables another: offers are deposited on the *th* names, which no line reads. The settling is therefore a finite, conflict-free sequence of eight steps up to the ordinary races for mark cells, and those commute because each restores its cell unchanged. Hence confluence and termination. □

Definition 11 (The settled modality)**.** $\langle\!\langle K_s \rangle\!\rangle\varphi$ holds when the claim at s fires and φ holds of the term after the settling of Lemma 10. This is the weak diamond, and Lemma 10 makes it deterministic in s.

We use $\langle\!\langle - \rangle\!\rangle$ throughout, and note that a plain $\langle K_s \rangle$ would land the scientist between a move and the world's recognition of it—a state in which the threat structure is stale. That the distinction has to be drawn at all is a fact about a world in which perception is a process rather than a lookup.

5 The ladder

5.1 Six rungs and a limit

The relaxation lattice for this domain has a ladder through it that every human learner of the game climbs, and it is convenient that the ladder is already named: it is essentially Newell and Simon's rule list. We write it in the generated logic, cumulatively, each rung adding one disjunct above the previous ones in priority.

rung	formula added	kind and depth
0	$\top$	—
1	$\exists s.\ \mathsf{Emp}(s) \wedge \mathsf{Threat}_s(\mathrm{me})$	spatial, depth 1
2	$\exists s.\ \mathsf{Emp}(s) \wedge \mathsf{Threat}_s(\mathrm{them})$	spatial, depth 1
3	$\exists s.\ \langle\!\langle K_s \rangle\!\rangle \mathsf{Fork}(\mathrm{me})$	one modal step over $\|_0$
4	$\exists s.\ \forall t.\ \langle\!\langle K_s \rangle\!\rangle [\![K_t]\!]\, \neg\mathsf{Fork}(\mathrm{them})$	box under diamond, depth 2
5	$\mathsf{Central}$	name-level, depth 0
6	$\chi := \nu X.\ \forall s.\ [\![K_s]\!]\big(\neg\mathsf{Lost} \wedge \exists t.\ \langle\!\langle K_t \rangle\!\rangle X\big)$	full alternation

Here $\mathsf{Emp}(s) := \mathsf{in}(sq_s)\top$—the square still offers a receipt, a structural predicate of depth one—and $\mathsf{Central}$ is a preference over square *names*, read off the reflected structure $@[*b, k]$ by a name predicate, containing no modality and requiring no step at all. The quantifiers $\exists s$ and $\forall t$ range over redex positions, which by Proposition 8 is to say over legal moves.

Two features of the shape are worth naming, and both were invisible while K was an uninstantiated placeholder.

Observation 12 (Winning and blocking are perception, not lookahead). Rungs 1 and 2 contain no modality. Having a winning move is not a fact about what would happen if one moved; it is the fact that a completing offer stands at a square still open, and the line processes deposited that offer when the world last settled. The scientist reads it. The behavioural content was pushed into the structure of the term by a computation somebody else already paid for.

This has a general form, and it is the most useful thing in the section. There are two routes to any predicate about the environment: *read* it, if the environment has already computed it into legible structure, or *drive* it, by taking steps and watching. The first is perception and the second interaction—grades one and two of Table 1 of [1]—and they are priced by different schedules, one by formula depth and one by rendezvous. An environment that computes a great deal about itself into structure is one in which a poor scientist can nonetheless know much; an environment that computes nothing until driven is one where knowledge is available only to those who can afford to act. That is a claim about environments rather than about learners, and this framework can state it because it prices the two grades separately.

Observation 13 (The asymmetry has moved to the right place). With K instantiated, the existential and universal quantifiers range over moves, and the ladder's one genuinely alternating rung is rung 4: $\exists s \forall t$, a box under a diamond. By Proposition 6 of [1] the diamond has a finite confirming witness and the box has none, so rung 4 is the rung whose confirmation is unbounded and whose refutation is cheap. Section 8 finds that it is also, by a wide margin, the worst purchase on the ladder: $+0.005$ of win probability for 111 metered units. Making a fork is establishing an existential; preventing one is discharging a universal, and the framework prices that difference before measuring it.

Remark 14 (Why χ is reachable here, and what that costs later). Proposition 9 of [1] says the complete theory of an environment is unaffordable when its recursion follows the reduction rather than a finite structural cycle. Noughts-and-crosses is the converse case of Remark 10 there: the board loses a square at every step, so the alternation in χ bottoms out and ν is exact at a finite unfolding—nine, in fact. This is a *closed* science, one of the subject matters admitting a complete finite theory. That is what makes the domain a good demonstration and, as §10 shows, what kills the ecology.

5.2 The rungs are the ν-approximants

The ladder is not merely ordered; it is metrically graded, and by the metric the companion note already fixed.

Proposition 15 (The ladder is a chain of shrinking steps). *Under the stratified distance of [1], Definition 8, the rungs agree on every approximant below the stage at which the added conjunct first looks and differ at that stage. Rungs 1 and 2 add conjuncts that inspect the settled term without stepping, and so sit at stage 1; rung 3 adds one $\langle\!\langle K_s \rangle\!\rangle$, reaching stage 2; rung 4 adds a box beneath it, reaching stage 3; and χ is the limit of the sequence. Hence $d(\varphi_r, \varphi_{r+1}) = 2^{-r}$ for $r = 1, \ldots, 4$, and the steps shrink as one climbs.*

Two consequences follow immediately and they are the ones that structure §9. First, moving from rung r to rung $r+1$ is a *short* move—distance 2^{-r}, shrinking as one deepens—so by confinement a scientist that only ever deepens can never leave the ball its early commitments placed it in. Second, by Proposition 14 of [1], cost is monotone in distance, so the cheap moves are exactly the confined ones.

5.3 The modal asymmetry, and what it predicts

Observation 13 located the ladder's one alternating rung. It is worth writing the asymmetry out, now that the quantifiers have something to range over:

$$\underbrace{\exists s.\ \langle\!\langle K_s \rangle\!\rangle \mathsf{Fork}(\mathrm{me})}_{\text{a finite witness: exhibit the square}} \qquad \text{versus} \qquad \underbrace{\exists s.\ \forall t.\ \langle\!\langle K_s \rangle\!\rangle [\![K_t]\!]\, \neg\mathsf{Fork}(\mathrm{them})}_{\text{no finite witness: every reply must be checked}} \quad .$$

Establishing that one *can* fork requires producing the move. Establishing that a move is *safe* requires discharging a universal over the opponent's replies, and by Proposition 6 of [1] no finite run witnesses that; only a counterexample is finite. The framework therefore predicts, from the shape of the formula rather than from anything about games, that offensive knowledge is the cheaper acquisition—which is what every human learner of the game in fact does, and which §8 confirms in the currency that matters here: rung 4 is the worst purchase on the ladder.

There is a second, independent measurement of the same asymmetry. Restricting the top heuristic rung to each colour separately: as the first player it achieves the game-theoretic value against a perfect opponent, drawing every game; as the second player it loses 6.3%. The same policy is complete on offence and incomplete on defence.

Remark 16 (A reading, offered as such). We resist over-claiming on the second measurement. The first player has the initiative, so existential reasoning is more nearly sufficient for it, and one could attribute the colour asymmetry to that alone. What the framework adds is a reason the two are not symmetric *in price*: the universal formula is the one whose confirmation has no finite witness, so a budget-bounded learner confirms attacks and can only ever refute claims of safety. We take the colour measurement as consistent with Proposition 6; the rung-4 cost is the demonstration.

6 The assay

A move is an experiment, and in this domain that is not a metaphor to be maintained but the only thing a move can be.

Definition 17 (The move-assay)**.** For a scientist holding hypothesis φ with metabolic channel m, a move on probe channel p is

```
contract Move(b, @hyp, @me, ret) = {
  new probe in {
      @[*b, "offer"]!(*probe, me) // the probe perturbs the world
    | for (@pos <- probe where pos |= hyp) { ret!(("confirm", pos)) }
    | for (@pos <- probe where ~(pos |= hyp)) { ret!(("refute", pos)) }
  }
}
```

The complementary pair is what makes this more than an expression of hope: in a concurrent setting a guard that simply fails to fire cannot be distinguished from a guard still waiting, so the negated guard is what separates refutation from patience. The three outcomes of Proposition 5 of [1] are three code paths, one of which is empty—and in this domain the empty path has a concrete reading. If the opponent does not reply within the budget, the game is abandoned unresolved; the scientist learns nothing and has paid for the attempt. That is $\perp$, and it is neither confirmation nor refutation.

Remark 18 (Where the checker is). Nothing in the listing consults a hypothesis and then decides. The hypothesis is in guard position, so the reduction rule is the checker, and the cost of testing is metered by the same apparatus that meters the move. An unpriced test would be an oracle and would collapse the motivational story entirely; this is the concrete form of that constraint.

6.1 Pricing

We charge each rule the depth of the formula it installs, times the number of candidate moves it must examine:

$$\kappa(d) = 2^d, \qquad \text{cost of a decision at rung } r \;=\; |M| \cdot \sum_{\text{rules } 1..r} \kappa(d_{\text{rule}}),$$

with $|M|$ the number of legal moves—which by Proposition 8 is the branching factor of the quantification over redex positions, so the two readings of $|M|$ agree. Rungs 1 and 2, being structural at depth 1, cost 2 per candidate each; rung 3 adds 4 for its single settled step over a separation; rung 4 adds 8 for the box beneath the diamond; $\mathsf{Central}$, being name-level at depth 0, adds 1; and χ costs 32.

Three comments. The schedule is a *choice*, and §8 varies it. The exponential shape is not arbitrary: it is what makes distance, depth, and cost one graded structure, which is the reason [1] prefers the ν-stratified ultrametric to a measure-theoretic alternative. And the positional rung costing almost nothing is not a thumb on the scale; it is a consequence of $\mathsf{Central}$ containing no modality, hence requiring no lookahead. That this cheap rung turns out to be the most profitable one on the ladder is the main empirical finding below, and it would have been invisible under a schedule that priced by rung index rather than by formula depth.

7 Destructive assay, again

Section 8 of [1] argues that because harvest ends the prey's computational life, each specimen affords exactly one measurement, so hypotheses about individuals are worthless and the hypothesis language must be a logic of namespaces. The argument is abstract there. Here it is the situation.

A scientist that wins eats its opponent, so it never faces that opponent again. It cannot accumulate a model of *this* player across games; what it can accumulate is a model of the *kind* of player—a namespace description, satisfied by many opponents, and the thing a name predicate $@\varphi$ states. Every rung of §5 is of this form: none of them mentions an individual, all of them describe a class of positions or a class of policies.

There is a corollary worth making explicit because it has a familiar shape.

Proposition 19 (Learning requires a population)**.** *In a world where victory is fatal to the loser, no single scientist can test a hypothesis about opponent policy more than once against the* *same opponent. Statistical confirmation therefore requires either many opponents drawn from one namespace, or a lineage that inherits the hypothesis. The wall supplies the first and §9 supplies the second.*

This is the same conclusion §17.2 of [1] reaches for continual learning by a different route: a population is required rather than merely preferred.

8 The measurements

Everything in this section is exact rather than sampled. The game has 5,478 reachable positions, 765 up to the symmetries of the square, and 255,168 distinct plays; of its 4,520 non-terminal positions, 1,890—41.8%— offer a fork to the player to move, which is why the rung-3 conjunct is worth having at all. Outcome distributions and expected metering are computed by recursion over the whole tree, with ties within a rung broken uniformly. Figures below are symmetrised over colour: each policy plays first half the time.

8.1 The ladder pays, and where

rung	added conjunct	win	draw	loss	cost	Δwin	return
0	$\top$	0.437	0.127	0.437	0.0	—	—
1	$\mathsf{Threat}_s(\mathrm{me})$	0.668	0.075	0.258	40.9	+0.231	+0.0057
2	$\mathsf{Threat}_s(\mathrm{them})$	0.798	0.165	0.037	77.8	+0.130	+0.0035
3	$\langle\!\langle K_s \rangle\!\rangle\mathsf{Fork}$	0.831	0.135	0.035	134.9	+0.033	+0.0006
4	$\forall t.[\![K_t]\!]\neg\mathsf{Fork}$	0.836	0.138	0.027	245.7	+0.005	+0.00005
5	$\mathsf{Central}$	**0.918**	0.080	0.002	255.9	+0.083	**+0.0082**
6	χ	0.873	0.127	**0.000**	679.2	−0.046	−0.0001

Table 1: The ladder against a uniform-random opponent. "cost" is expected metering per game under $\kappa(d) = 2^d$; "return" is Δwin per metered unit. Loss rate falls monotonically along the whole ladder. Win rate does not: it peaks at rung 5.

Three features of Table 1 deserve separate statements.

Observation 20 (Diminishing returns, until they don't). Marginal return falls by more than two orders of magnitude from rung 1 to rung 4—+0.0057, +0.0035, +0.0006, +0.00005—while cost roughly doubles at each step. Rung 4 in particular buys +0.005 of win probability for 111 metered units, and it is the alternating rung of Observation 13: the one whose formula has no finite confirming witness. A budget-bounded learner would stop before it, and the foraging inequality of [1], Proposition 27, says exactly when: harvest credits the scientist only if the prey's stack exceeds what finding and opening it cost.

Observation 21 (The non-modal rungs carry the ladder). Once K is instantiated, three of the five useful rungs turn out to contain no modality: rungs 1 and 2, which read standing offers, and rung 5, which reads square names. Together they deliver +0.444 of the ladder's total +0.481 gain —92.3% of the yield—for 87.9 of its 255.9 metered units, or 34.3% of the cost. The two modal rungs deliver the remaining +0.038 for 168.0 units. Rung 5 alone returns +0.0082 per unit, the best figure on the ladder and some one hundred and sixty times rung 4's.

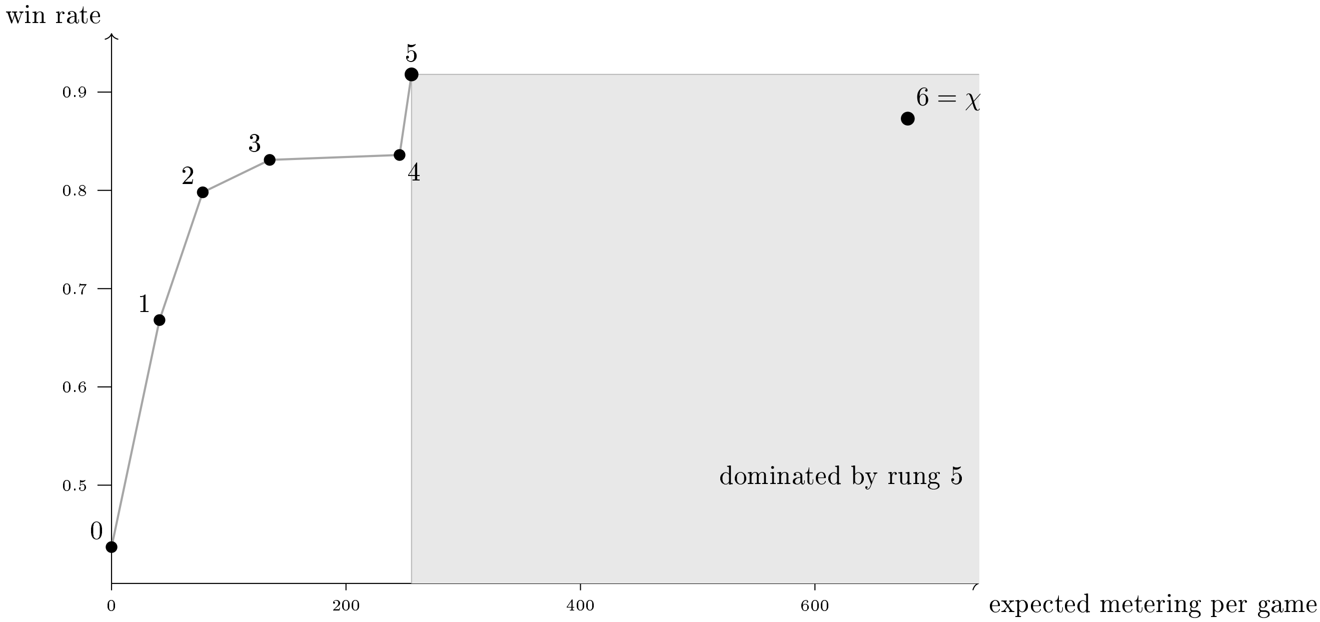


Figure 2: The ladder as a cost–yield curve, against a uniform-random opponent. Rungs 1–4 show sharply diminishing returns for a roughly doubling price, rung 4 —the alternating one—being the flattest. Rung 5, a predicate on square names of depth zero, is nearly free and delivers the largest jump on the ladder. The complete theory χ lies inside the shaded region: it costs more than rung 5 and wins less, so it is dominated on both axes.

This is the result we would lead with, and it is one we could not have stated before the correction: with $\langle K \rangle$ standing in as a placeholder, rungs 1 and 2 looked modal, and the split between reading and driving was invisible. The classifier note argues, in the knot setting, that the spatial layer is where the generated logic earns its name and that a purely behavioural logic cannot state the properties that matter ([2], Remark 5). That argument is made there on expressiveness grounds. Here it is cashed in an ecological currency: the non-behavioural fragment is not merely expressible-and-useful, it is where nearly all the yield per token lives, and a mortal learner that priced its own hypotheses would find it first. A modal logic could not have offered any of it.

Remark 22 (Why this is not an artefact of the encoding). One might object that we simply put the threats into the term, so of course reading them is cheap. The objection has force and the answer is that somebody paid: the line processes run on every settling, and §6.1 charges that scan. What the encoding does is *relocate* the cost, from the scientist's budget to the world's—and that relocation is precisely the distinction between an environment one can perceive and one that must be driven. The interesting quantity is not whether the work is done but who is billed for it, which is a question this framework can pose because both parties have stacks.

Observation 23 (The true theory is dominated). χ is the complete theory: it never loses, from either colour, against any opponent. It is also, against fallible prey, strictly worse than the heuristic rung below it on both axes at once. Against a random opponent rung 5 wins 0.918 at cost 255.9 while χ wins 0.873 at cost 679.2; against a rung-4 opponent, 0.448 at 280.3 versus 0.418 at 708.8. Fewer wins, and between one and a half and two and a half times the price.

The mechanism is not mysterious and is well known to anyone who has implemented game search: minimax assumes optimal opposition, so it settles for a draw in positions where a heuristic would set a trap that a fallible opponent walks into. What is worth noting is that the framework *predicted* this before we measured it. Remark 16 of [1] says that a mortal scientist is not rewarded for holding true hypotheses but for holding nutritive ones, and that a lineage under selection drifts toward yield-biased search wherever yield and discrimination disagree. Here they disagree, the disagreement is measurable, and it runs in the predicted direction.

Corollary 24 (Truth buys safety, not food)**.** *χ is the unique rung with loss probability zero. So the complete theory does buy something the heuristic does not—it cannot be beaten—but what it buys is insurance rather than yield. In a world where survival is the only score and the prize is another's stack, insurance is worth having only when the stakes are large enough to matter, which is the next calculation.*

8.2 Invasion, and the price of being right

Take a population fixed at rung 5 and ask when a χ-holding mutant invades. Head to head, χ wins 3.17% and never loses; it costs 720.0 per game against the resident's 324.0. The mutant is favoured when $\sigma \cdot 0.0317 > \text{price} \cdot 396.0$, that is when

$$\sigma \ > \ 12474 \cdot \text{price},$$

where σ is the prey stack won and *price* converts metered units into tokens. The complete theory is therefore a luxury good, purchasable only where the stakes are some twelve thousand times the unit cost of thought.

Remark 25 (Basic research, priced). Remark 80 of [1] concedes the pragmatist's objection—this scientist's epistemology is metabolic and truths that do not pay go unfunded—and answers with surplus: a scientist holding more than its foraging horizon requires can afford inquiry that does not pay. The number above is that remark made quantitative for one domain. Truth is affordable here, and only here, above a threshold; below it, the ecology selects for a theory that is known to be wrong and known to be more profitable. Pricing the ladder by the instantiated formulae rather than by rung index raises that threshold by a factor of three, because the cheap rungs turn out to be cheaper than a placeholder $\langle K \rangle$ made them look.

8.3 The phase band

Table 2 reports, for each opponent depth and each prey stack, which rung maximises $\sigma \cdot P(\text{win}) - \text{price} \cdot \mathbb{E}[\text{cost}]$.

The shape is the phase diagram of [1], §13.2, in a domain where both order parameters have concrete readings.

- **Low δ, low stakes: theory is overhead.** Against a shallow opponent worth little, rung 0 or 1 is optimal. Reflex beats inquiry; a theory is something one is paying to carry.
- **High δ: inquiry cannot pay.** Against a rung-6 opponent the optimal rung is 0 at *every* stake, because no rung ever beats it, so the cheapest available policy maximises net. This is the upper boundary, and it has a bracing reading: against an environment one cannot model within any affordable budget, the correct scientific policy is not to try.

σ	vs r0	vs r1	vs r2	vs r3	vs r4	vs r5	vs r6
20	2	2	0	0	0	0	0
50	2	2	3	3	5	0	0
100	5	5	5	5	5	0	0
200	5	5	5	5	5	2	0
400	5	5	5	5	5	2	0
800	5	5	5	5	5	2	0
1600	5	5	**6**	**6**	5	2	0

Table 2: Optimal rung, at price 0.05 per metered unit, as a function of the prey's stack σ (rows) and the prey's own rung (columns). The optimal rung is 0 in both extremes and interior in between.

- **The band between.** For intermediate opponents and stakes, an interior rung is optimal—mostly rung 5, the heuristic, not rung 6, the truth.

χ is optimal in exactly two cells of the whole grid, both at the cheapest price and the largest stake. Raising the price to 0.2 and to 1.0 shifts the band toward the cheap rungs without changing its shape: at price 1.0 the optimal rung is 0 over most of the table and never 6 anywhere. The qualitative structure—interior optimum, degenerate at both ends—is robust; the boundaries are not.

9 Confinement, measured

This section is the one we would keep if we could keep only one, because it turns Proposition 12 of [1] from a fact about ultrametric spaces into a fact about a hypothesis space someone might actually hold.

9.1 Two loci, and only one of them is reachable

A policy on this ladder has exactly two parameters, and they are the two that Definition 13 of [1] says a revision move must choose: a *locus* and an *operation*. The loci here are

- the **opening**—which square to take first—which is the stage-1 approximant, at metric distance 2^{-1}; and
- the **rung**—how deep the theory runs—which is everything at stage 2 and below.

Deepening the rung is a short move: from rung r to rung $r+1$ is a step of 2^{-r}. Changing the opening is a long one. By confinement, a scientist whose revisions are all deepenings can never change its opening, however long it lives and however many refutations it suffers. In the listing of Appendix A this is visible as a fact about the code: `Revise` only ever increments the rung.

9.2 The measurement

Lock the opening and let both sides otherwise play the top heuristic rung. Then against a rung-2 opponent:

opening	win rate at rung 5	win rate at rung 6 (χ)
centre	**0.833**	0.548
corner	0.597	**0.896**
side	0.318	0.710

Read the two columns against each other, because the disagreement is the point.

Observation 26 (The best opening is not a property of the game). Under the heuristic the centre is much the best opening; under the complete theory the corner is, by a wide margin, and the centre is the *worst* of the three. The stage-1 commitment cannot be evaluated in isolation from the stages below it, and the ordering reverses.

Corollary 27 (Deepening within a ball can make things worse). *A lineage committed to the centre opening that upgrades its theory from rung 5 to χ falls from* 0.833 *to* 0.548*. Every step of that upgrade is a well-motivated refinement, each is licensed by refutations, and the result is a worse forager. Improvement at depth is not improvement, when the shallow commitment was wrong for the deep theory.*

This is Proposition 12 and Proposition 14 acting together and it is worth being precise about which does what. Confinement says the walk cannot reach the corner opening. Cost-monotonicity says the move that would reach it is the expensive one, because it invalidates every confirmation obtained at depth ≥ 1—which is all of them. The scientist is trapped by geometry and priced out of the escape by its own history of successful assays.

9.3 The escape, and what it costs whom

§6.5 of [1] says an individual has exactly one route out and it is a poor one, and that recombination is the other. Here is that claim, measured.

Consider a mutant identical to the resident population in every respect—same rung, same rules, same metering—differing only in the opening allele, which is a corner rather than the centre. It arises by crossover at the shallow locus: the opening allele of one parent with the rung allele of another.

	win	draw	loss
mutant as first player, vs resident	0.333	0.667	0.000
mutant as second player, vs resident	0.000	1.000	0.000
resident vs resident	0.000	1.000	0.000

The mutant wins a third of the games in which it moves first, never loses, and costs exactly what the resident costs, since it is the same policy with one different move. It invades unconditionally, at any stake, at any price.

Proposition 28 (The invading move is the one no individual can make). *The corner-opening mutant differs from the resident by a revision of distance 2^{-1}. By Proposition 12 no walk of deepenings reaches it; by Proposition 14 its cost to an individual is the invalidation of every confirmation the individual holds. It is nonetheless free to the lineage, because the cost falls on an offspring that does not yet exist and has no confirmations to invalidate.*

That is the whole argument of §12 of [1] for why reproduction is part of the epistemology and not an appendix to it, exhibited on a hypothesis space small enough to check exhaustively. Sex is not here a mechanism for generating variation in general; it is a mechanism for making *shallow* revisions, which are exactly the ones the metric denies to a living scientist.

Remark 29 (Opening theory changes by catastrophe). The prediction generalises, and it is testable outside this framework: in any domain whose hypothesis space is ultrametric, early commitments should change discontinuously—by generational replacement, by schism, by the arrival of an outsider—and not by incremental refinement, whereas late commitments should change smoothly and continuously. That is a recognisable description of how opening theory in real games, and paradigms in real sciences, actually move. We note the resemblance and claim nothing further from it.

10 Succession, and the death of a solved world

Noughts-and-crosses is a draw under optimal play. That fact, harmless in game theory, is fatal here.

Proposition 30 (A solved science is a dead ecology). *In a conserved world whose only predatory transfer is the arena's, a population all of whose members hold theories that draw against one another performs no harvest. Its members continue to burn tokens to live, and the wall's reserve is finite. Hence σ declines monotonically to zero and the population starves, converging faster the deeper its theories, since burn rate rises with the rung held.*

Both the χ monoculture and the centre-opening rung-5 monoculture are of exactly this kind: we measure all draws, harvest rate 0.000, in both. The better the population's science, the sooner it dies.

Corollary 31 (Imperfection is load-bearing). *The corner-opening rung-5 population, by contrast, sustains a win rate of* 0.333 *for the first player against itself. Harvest continues; tokens circulate; the ecology persists. A strictly* worse-*informed population is the living one.*

We resist reading a moral into this, but the structure is worth stating plainly because it is a consequence of conservation and not of anything we chose. Predation is the purchase of somebody else's remaining time ([1], Remark 19). A population that has solved its environment has nothing left to learn about each other and no asymmetry to exploit, so the only remaining transfer is metabolic burn, which is one-way into the record. Science, in this world, is a phase; and the far side of the phase boundary is not error but exhaustion.

Remark 32 (What would keep it alive). Three exits, none of which this note takes. Enlarge the game so that χ is unaffordable, putting the domain back on the Proposition 9 side of Remark 10— the recursion following the reduction rather than the structure. Let the arena's rules drift, so the environment is non-stationary and χ has a moving target. Or admit a genuine external source that is not exhaustible on the timescale of the population, which is to say, redraw the boundary of the conserved system. The third is the one biology took, and [1], §11.1 argues it is a bookkeeping choice about where the cut lies rather than a change in the physics.

11 What this shows, and what it does not

Shown. The generated logic contains the game's strategic vocabulary as terms rather than as hand-coded features, and the requirement that it do so constrains the encoding of the world (§4). Pricing hypotheses by formula depth yields a cost–benefit structure whose shape is neither monotone nor obvious, in which the spatial conjunct is the most profitable purchase on the ladder and the complete theory is dominated (§8). Confinement is not an abstract property of ultrametrics but a specific trap on this ladder, with a specific escape available only to a lineage (§9). And conservation forces an ending (§10).

Not shown. We closed one of the companion note's declared free parameters—the revision policy—by adopting its own default, refutation rate as temperature, and a different policy would give different numbers, though we expect not a different shape. We did not implement the ideal guards: the structural and modal formulae of §5 are specifications, and the running listing checks their arithmetic shadows. The pricing schedule is a choice. And the ecology of Appendix A is written but not run to a fixed point; the measurements of §8 are exact game-tree computations over policies, not observations of a population, so every claim about invasion is a claim about payoffs, not a simulation result.

The honest summary. What the exercise establishes is that the framework is *contentful*: instantiating it on a known domain produces statements that could have come out otherwise and did not have to come out as the companion note predicted. That the spatial rung would be the best buy, that χ would be dominated, and that the invading mutant would be a shallow one were all consequences we computed rather than results we arranged.

12 Open problems

1. **Run the ecology.** Appendix A specifies a population; §8 computes payoffs. Closing the gap—sampling trajectories via the Gillespie construction of [6]—would turn Table 2 from a payoff calculation into the ensemble test that is the first open problem of [1].
2. **An unsolvable game.** Everything interesting about §10 follows from the domain being closed. Rerunning the mapping on a game whose χ is unaffordable—Connect Four is solved but not within a plausible budget; Hex is not solved at all—would test whether the phase band persists when the upper boundary is set by affordability rather than by the opponent's perfection.
3. **Is the ladder the lattice?** We wrote a ladder through the relaxation lattice, guided by what human players learn. Whether the lattice's own structure—the order in which forgetting conjuncts of χ yields satisfiable weakenings—recovers approximately this ladder, or a different one, is checkable for this game and would say whether "what people learn" and "what the logic makes cheap" agree.
4. **The reversal.** §9 finds that the optimal opening under χ and under the heuristic are different squares. Is that an artefact of this game, or is there a general statement—that the optimal shallow commitment is a function of the depth of the theory it heads, so that shallow and deep revisions cannot be separated? If the latter, it sharpens confinement considerably.

5. **Co-learning.** When both players revise, each is a moving target for the other and Table 2 is a snapshot of a game whose payoffs are themselves changing. This is problem 3 of [1] and this domain is small enough to attack it in.

A The world, in rholang

The listing below is the whole construction. Guards using structural or modal connectives are marked `//! ideal` and are specifications rather than checks the current reducer performs; everything else runs. Conservation can be audited by inspection: `Wall`, `Harvest`, and `Beget` move tokens, `Live` debits them from σ into κ, and nothing mints.

```
// ==========================================================================
// A WORLD OF MORTAL SCIENTISTS THAT PLAY NOUGHTS-AND-CROSSES
// ==========================================================================
//
// Companion listing to "Learning to Play". Every construction of the
// mortal-scientist note appears here as ordinary code:
//
// Sec. 3 the arena -- Arena, Turn, Harvest (the admitted context class)
// Sec. 4 the board -- Board, Cells, Square, Line, Scan
// Sec. 4.3 the context K_s -- Square, Choose (a ply is ONE rewrite)
// Sec. 5 the ladder -- Ladder (hypotheses as formulae)
// Sec. 6 the assay -- Move (a move IS an experiment)
// Sec. 6.1 metabolism -- Live, Wall
// Sec. 9 the two loci -- Beget, Crossover (opening x rung)
// Sec. 10 the world -- the bottom block
//
// CONVENTION. A parameter written `c` is a NAME (a channel); a parameter
// written `@d` is DATA. To pass a channel one sends `*c`, since @(*c) = c.
//
// NOTATION. Guards are written in the generated spatial-behavioural logic:
// out(n)phi, in(n)phi, phi | psi, the graded separation phi |_k psi, the
// hidden-name quantifier H x. phi, the name predicate @phi, the context-
// labelled modality <K>phi, and the greatest fixed point nu X. phi. The
// current interpreter accepts only the arithmetic and pattern fragment of
// `where`; guards using a structural or modal connective are marked
// `//! ideal` and are specifications of a check rather than a check the
// reducer performs. Everything else runs.
//
// CONSERVATION. Nothing here mints. Wall moves reserve into a player;
// Harvest moves a loser's stack into a winner's; Beget moves a parent's stack
// into a child's; every Live debit moves a token from sigma into kappa. So
// sigma + kappa = Theta throughout, with Theta the wall's opening reserve plus
// the players' opening stacks.
// ==========================================================================

new stdout(`rho:io:stdout`), display, arena, wall, mate, ladder,
    Live, Wall, Board, Cells, Square, Line, Scan, Arena, Turn, Adjudicate,
    Harvest, Choose, Positions, Move, Revise, Ladder, Pick, Crossover, Beget,
    Body, PlayOne, Mate
in {

  for (@line <= display) { stdout!(line) }

  // ========================================================================
```

```
// METABOLISM -- reclaim, debit, reissue. (Companion note, Def. 20.)
// Between the receipt and the send the stack is a message in flight: living
// is racing others to your own stack, and the metabolic channel m is exactly
// the vulnerability that Harvest exploits.
// ============================================================================

| contract Live(m, @burn, step) = {
    for (@sigma <- m) {
      match sigma > burn {
        true => { m!(sigma - burn) | *step | Live!(*m, burn, *step) }
        false => { display!(("DIED", "could not fund a step")) }
      }
    }
  }

// ============================================================================
// THE PRACTICE WALL -- a source, not a prey.
// Rate-limited (one contract), non-exclusive (anyone may call it),
// non-adaptive (its policy never revises), and finite. That access profile
// is what makes it a SOURCE in the sense of the companion note, Sec. 11.2,
// and what makes this world mixotrophic rather than purely predatory.
// ============================================================================

| contract Wall(w, @quantum, @reserve) = {
    for (ret <- w) {
      match reserve >= quantum {
        true => { ret!(quantum) | Wall!(*w, quantum, reserve - quantum) }
        false => { ret!(0) | Wall!(*w, quantum, 0) } // exhausted
      }
    }
  }

// ============================================================================
// THE BOARD -- and why a ply must be exactly one rewrite.
//
// Square k of board b carries TWO reflected names: a MOVE CHANNEL
// sq_k = @[*b,k] and a MARK CELL mk_k = @[*b,"m",k]. An EMPTY square is a
// RECEIPT offering to be claimed; an OCCUPIED square is that receipt gone
// and a resting send on its mark cell.
//
// This is what makes a ply a single rewrite, and that is what makes <K_s> a
// GENERATED modality rather than a derived abbreviation: the redex position
// of the claim at s IS the move at s, and the one-hole context K_s is the
// whole rest of the board (Def. 5, Prop. 6). Nine empty squares are nine
// redex positions, so the branching factor of ‘exists s’ is the legal move
// count.
//
// Offers live on a THIRD family, th_s = @[*b,"t",s], and not on the move
// channels. Were a line to signal a threat by sending on sq_s, the signal
// would rendezvous with the square’s own receipt and the line would have
// PLAYED THE MOVE. Perception must not be able to act; that separation of
// grades one and two is enforced here by a choice of namespace.
// ============================================================================

| contract Board(b, @marks) = { Cells!(*b, marks, 0) | Scan!(*b) }

| contract Cells(b, @marks, @k) = {
```

```
    match k < 9 {
      true => {
          @[*b,"m",k]!(marks.nth(k)) // the mark cell
        | match marks.nth(k) == 0 {
            true => { Square!(*b, k) } // empty: offer a receipt
            false => { Nil } // occupied: no receipt
          }
        | Cells!(*b, marks, k + 1)
      }
      false => { Nil }
    }
  }

// An empty square. The claim is ONE rewrite: a join consuming the mover's
// send and the stale mark together. After it fires no receipt remains, so the
// square is no longer a redex position -- which is exactly Prop. 6.
| contract Square(b, @k) = {
    for (@p <- @[*b,k] & @old <- @[*b,"m",k]) { @[*b,"m",k]!(p) }
  }

// A line reads its three MARK CELLS and replaces them; it never touches a
// move channel. When it finds two marks of one player and one empty square s
// it deposits a standing offer on th_s. A threat is therefore not computed on
// demand: it is a piece of the term, and so visible to the structural layer.
| contract Line(b, @i, @j, @k) = {
    for (@x <- @[*b,"m",i] & @y <- @[*b,"m",j] & @z <- @[*b,"m",k]) {
        @[*b,"m",i]!(x) | @[*b,"m",j]!(y) | @[*b,"m",k]!(z)
      | match (x, y, z) {
          (p, q, 0) where p == q and p > 0 => { @[*b,"t",k]!(p) }
          (p, 0, q) where p == q and p > 0 => { @[*b,"t",j]!(p) }
          (0, p, q) where p == q and p > 0 => { @[*b,"t",i]!(p) }
          _ => { Nil }
        }
    }
  }

// The eight lines, re-run after each claim. This is the SETTLING of Lem. 8:
// each line fires once, no line's output enables another (offers go to the
// th names, which no line reads), so it terminates and is confluent -- which
// is what makes the settled modality <<K_s>> deterministic in s.
// Scanning is metered like everything else: perception is priced.
| contract Scan(b) = {
      Line!(*b,0,1,2) | Line!(*b,3,4,5) | Line!(*b,6,7,8)
    | Line!(*b,0,3,6) | Line!(*b,1,4,7) | Line!(*b,2,5,8)
    | Line!(*b,0,4,8) | Line!(*b,2,4,6)
  }

// ===========================================================================
// THE LADDER -- six hypotheses about the position, and chi.
//
// Each rung is a formula in the generated logic; the rung index is the modal
// depth of the conjunct it adds, hence by Def. 8 of the companion note the
// ultrametric stage: d(Phi_r, Phi_{r+1}) = 2^-r.
//
// Threat(p) := H s. out(s)@("claim", p) | TOP spatial, depth 1
// Fork(p) := Threat(p) |_0 Threat(p) spatial, k = 0
// Win(p) := <K> out(over)@("won", p) modal, depth 1
```

```
// Central := a pure preference over squares spatial, depth 0
//
// Note the shape of rungs 1 and 2. Win is existential and the safety
// conjunct universal, so by Prop. 6 of the companion note Win is finitely
// confirmable while safety is only finitely refutable: attack is cheap to
// establish, defence cheap only to destroy.
// ==========================================================================

//! ideal -- cumulative, highest priority first.
//
// Emp(s) := in(sq_s)TOP the square still offers a receipt
// Thr_s(p) := out(th_s)@p a completing offer stands at s
// Fork(p) := (H s. Thr_s(p) | TOP) |_0 (H s. Thr_s(p) | TOP)
// <<K_s>>phi := the claim at s fires, the board settles (Lem. 8), phi holds
//
// ‘exists s’ and ‘forall t’ range over REDEX POSITIONS, which by Prop. 6 is
// to say over legal moves. Note where the alternation is: rungs 1 and 2 have
// no modality at all -- winning and blocking are PERCEPTION, reading offers
// the last settling deposited -- and the one genuinely alternating rung is 4,
// a box under a diamond. By Prop. 6 of the companion note that is the rung
// with no finite confirming witness, and Sec. 8 measures that it is also the
// worst purchase on the ladder.
| contract Ladder(@r, ret) = {
   match r {
     0 => { ret!( @{ TOP } ) }
     1 => { ret!( @{ exists s. Emp(s) and Thr(s, me) } ) }
     2 => { ret!( @{ exists s. Emp(s) and (Thr(s, me) or Thr(s, them)) } ) }
     3 => { ret!( @{ exists s. Emp(s) and (Thr(s, me) or Thr(s, them))
                                 or <<K s>> Fork(me) } ) }
     4 => { ret!( @{ exists s. Emp(s) and (Thr(s, me) or Thr(s, them))
                                 or <<K s>> Fork(me)
                                 or forall t. <<K s>>[K t] ~Fork(them) } ) }
     5 => { ret!( @{ exists s. Emp(s) and (Thr(s, me) or Thr(s, them))
                                 or <<K s>> Fork(me)
                                 or forall t. <<K s>>[K t] ~Fork(them)
                                 or Central(s) } ) }
     // chi: exact at a finite unfolding, because the board loses a square at
     // every step, so the alternation bottoms out at nine. This is the
     // CLOSED-science case (companion note, Rem. 10) -- and Sec. 10 is where
     // that turns out to be fatal.
     6 => { ret!( @{ nu X. forall s. [K s]( ~Lost and exists t. <<K t>> X ) } ) }
   }
  }

// ==========================================================================
// THE ASSAY -- a move is an experiment.
//
// The complementary guard pair of Def. 4, verbatim. The hypothesis is not
// consulted and then acted on: it sits in guard position, so playing the
// move IS testing it and confirmation IS the rendezvous. The three outcomes
// of Prop. 5 are three code paths, and the third is empty -- if the world
// does not answer within budget the scientist has paid and learned nothing,
// which is ignorance, not refutation.
// ==========================================================================

// Choosing WHICH redex position to fire is the whole of the policy. The
//! ideal guard ranges over the empty squares -- that is, over the redex
```

```
// positions of Def. 5 -- and picks the first satisfying the held rung. By
// Prop. 17 of the companion note the number it can afford to evaluate is its
// stack divided by the cost of one check, so a poor scientist does not survey
// its own branching.
| contract Choose(b, @hyp, @me, ret) = {
   new cand in {
       Positions!(*b, *cand, 0)
     | for (@s <- cand where Sat(s, hyp, me)) { //! ideal
         new r in { Move!(*b, s, hyp, me, *r)
                  | for (@o <- r) { ret!(s, o) } }
       }
   }
  }

| contract Positions(b, cand, @k) = {
   match k < 9 {
     true => { cand!(k) | Positions!(*b, *cand, k + 1) }
     false => { Nil }
   }
  }

| contract Move(b, @s, @hyp, @me, ret) = {
   new probe in {
       @[*b, s]!(me) // THE CLAIM: one rewrite, at redex s
     | Scan!(*b) // the settling of Lem. 8
     | @[*b,"state"]!(*probe)
                                              //! ideal: structural guards
     | for (@pos <- probe where pos |= hyp) { ret!(("confirm", pos)) }
     | for (@pos <- probe where ~(pos |= hyp)) { ret!(("refute", pos)) }
   }
  }

// Revision walks the lattice. The move has the two parameters the metric
// grades (Def. 13): a LOCUS and a DIRECTION. Here refutation rate is the
// temperature, which is the companion note's own default policy -- and it is
// an observable the assays already emit, not a schedule imposed from outside.
| contract Revise(@rung, @refutes, @confirms, ret) = {
   match refutes * 4 > confirms {
     true => { ret!(rung + 1) } // deepen: a move of distance 2^-rung
     false => { ret!(rung) }
   }
   // NB no case touches the OPENING. By Prop. 12 that is not an oversight:
   // the opening is the stage-1 approximant, a move of distance 2^-1, and an
   // individual can neither reach it by deepening nor afford it if it could.
   // Sec. 9 is where the opening changes, and it is not here.
  }

// ==========================================================================
// THE ARENA -- the rules, holding no stack of its own.
//
// On a win the arena hands the winner the loser's metabolic NAME. It mints
// nothing; it ROUTES, which is what keeps Sec. 16 of the companion note
// intact -- an exogenous reward would need a referee with a mint, and a
// referee with a mint destroys conservation.
//
// In the vocabulary of Def. 25 the arena IS the choice of admitted context
// class A: the contexts it admits to hold m_P are exactly those that have
```

```
// completed a line against P, and by Thm. 24 those are non-image contexts.
// Predation is legal here for precisely one reason, and it is that somebody
// won.
// ============================================================================

| contract Arena(@hX, mX, @hO, mO, ret) = {
   new b, over in {
       Board!(*b, [0,0,0,0,0,0,0,0,0])
     | Turn!(*b, hX, *mX, hO, *mO, *over, 1)
     | for (@result <- over) {
         match result {
           ("won", 1) => { Harvest!(*mO, *mX, *ret) } // crosses eat noughts
           ("won", 2) => { Harvest!(*mX, *mO, *ret) }
           "draw" => { ret!("draw") } // nobody eats
         }
       }
   }
  }

| contract Turn(b, @hX, mX, @hO, mO, over, @who) = {
   new played in {
       Choose!(*b, if (who == 1) { hX } else { hO }, who, *played)
     | for (@s, @outcome <- played) {
         Adjudicate!(*b, *over, who)
       | Turn!(*b, hX, *mX, hO, *mO, *over, 3 - who)
     }
   }
  }

//! ideal -- the win test is a modal formula, not a scan of a list
| contract Adjudicate(b, over, @who) = {
   new r in {
       @[*b, "state"]!(*r)
     | for (@pos <- r where pos |= @{ Line3(who) }) {
         over!(("won", who))
       }
     | for (@pos <- r where pos |= @{ ~Line3(who) and
                                   ~ exists s. Emp(s) }) {
         over!("draw")
       }
   }
  }

// Harvest (Def. 22): an ordinary receipt on the loser's metabolic channel.
// No new rewrite rule is needed. The loser dies because the message it was
// going to reclaim has been consumed.
| contract Harvest(mPrey, mPredator, ret) = {
   for (@sigma <- mPrey) {
     for (@own <- mPredator) {
         mPredator!(own + sigma)
       | ret!(("harvest", sigma))
       | display!(("HARVEST", sigma, "tokens"))
     }
   }
  }

// ============================================================================
```

```
// REPRODUCTION -- two loci, and why that is the whole point.
//
// The genome is a quotation: inert, transmissible, inspectable, and free
// until dropped. Two loci are exposed and they are not arbitrary -- they are
// the two parameters of Def. 13. The RUNG allele is the depth of the theory;
// the OPENING allele is the stage-1 commitment.
//
// Crossover exchanges them independently, and a swap at the opening locus is
// a move of distance 2^-1 -- exactly the move Prop. 12 forbids to any
// individual walking incrementally. It is free to the lineage because the
// cost falls on an offspring that has no confirmations to invalidate. This
// is the escape of Sec. 6.5, and Sec. 9 measures that it invades.
// ==========================================================================

| contract Pick(@a, @b, @bit, ret) = {
    match bit { 0 => { ret!(a) } 1 => { ret!(b) } }
  }

| contract Crossover(@mother, @father, @chi, ret) = {
    match (mother, father, chi) {
      ([om, rm], [of, rf], [b1, b2]) => {
        new r1, r2 in {
            Pick!(om, of, b1, *r1) // the opening locus -- stage 1
          | Pick!(rm, rf, b2, *r2) // the rung locus -- deeper
          | for (@o <- r1 & @r <- r2) { ret!([o, r]) }
        }
      }
    }
  }

// Birth. The child's code is CONSTRUCTED as a name and rests inert on `code`:
// it does not reduce, does not age, and burns nothing. Dropping it with * is
// what makes it a phenotype and what starts the meter. The guard enforces the
// minimum viable endowment -- a parent may not endow below what a child needs
// to fund a first game.
| contract Beget(m, @alleles, @endow, ret) = {
    for (@sigma <- m where sigma > endow + 40) {
        m!(sigma - endow)
      | match alleles {
          [o, r] => {
            new childM, code in {
                code!( @{ Body!(*childM, o, r, endow) } ) // the germ: inert
              | for (g <- code) { *g | childM!(endow) } // the soma: metered
              | ret!(("born", o, r))
            }
          }
        }
    }
  }

// ==========================================================================
// THE SCIENTIST -- a metered loop of games and matings.
// Alleles vary; this does not.
// ==========================================================================

| contract Body(m, @opening, @rung, @endow) = {
    new hyp, score in {
```

```
        score!(0, 0)
      | Ladder!(rung, *hyp)
        // The burn rate is the cost of HOLDING the theory, paid before any
        // assay is run. It is why the top rung is not free, and why Sec. 8
        // finds the exact theory chi dominated by the rung beneath it.
      | Live!(*m, 2 + rung, @{
            PlayOne!(*m, opening, *hyp, *score)
          | Mate!(*m, [opening, rung], endow)
        })
    }
  }

| contract PlayOne(m, @opening, hyp, score) = {
    new ret in {
        for (@h <- hyp) { hyp!(h) | Arena!(h, *m, "wall", *wall, *ret) }
      | for (@outcome <- ret) {
          for (@c, @f <- score) {
            match outcome {
              ("harvest", _) => { score!(c + 1, f) }
              "draw" => { score!(c, f) }
              _ => { score!(c, f + 1) }
            }
          }
        }
    }
  }

// Mating requires DISCLOSURE: the genome goes out on a public channel where
// anything able to read it can. Nothing declares a species; the `where` guard
// is the only thing keeping two lineages apart, and deleting it hybridises
// them.
| contract Mate(m, @alleles, @endow) = {
    mate!(alleles)
  | for (@partner <- mate where Conspecific(partner, alleles)) { //! ideal
      new ret in {
          Crossover!(alleles, partner, [0, 1], *ret)
        | for (@kid <- ret) { Beget!(*m, kid, endow, *display) }
      }
    }
  }

// ===========================================================================
// THE WORLD
// Theta = 12000 (wall reserve) + 4 x 400 (players) = 13600 tokens. No more.
// ===========================================================================

| Wall!(*wall, 6, 12000)

// Four lineages, differing only at the two loci. The pair that shares a rung
// and differs at the opening is the whole experiment: Sec. 9 measures that
// the corner opener strictly invades the centre opener at IDENTICAL metering
// cost, and that no amount of deepening lets the centre opener reply.
| new a1, a2, b1, b2 in {
      a1!(400) | Body!(*a1, 4, 5, 120) // centre opening, top heuristic
    | a2!(400) | Body!(*a2, 4, 6, 120) // centre opening, the exact theory
    | b1!(400) | Body!(*b1, 0, 5, 120) // corner opening, top heuristic
    | b2!(400) | Body!(*b2, 0, 2, 120) // corner opening, shallow theory
```

```
    }

  // ------------------------------------------------------------------------
  // WHAT TO WATCH
  //
  // * Conservation is checkable by inspection. Wall, Harvest and Beget move
  // tokens; Live debits them. Nothing mints.
  //
  // * The hypothesis is never consulted. It is a guard. No line of code
  // evaluates a theory and then decides -- deciding is the comm.
  //
  // * Perception is priced twice over: Scan runs after every move, and the
  // burn rate rises with the rung held. Both are the same schedule.
  //
  // * Nobody screens an embryo. Crossover produces a genome and Beget drops
  // it. A parent could model-check the recombinant against a viability
  // formula and does not, because the check is priced against the same
  // stack -- selection is post hoc because screening costs.
  //
  // * No scientist revises its opening. Revise only ever increments the rung.
  // The opening changes exactly once per lineage, at Crossover, and free.
  //
  // * a2 holds the TRUE theory and is the one to watch starve. It burns 8 per
  // step against b1's 7, wins no more often against fallible opponents, and
  // draws every game against a1. Truth buys it safety and no food.
  //
  // * And it ends. When the wall's reserve reaches zero the only credit left
  // is another player's stack; when the survivors converge on theories that
  // draw against one another, Arena returns "draw", Harvest never fires, and
  // the remaining tokens sit in metabolic channels until the burn takes
  // them. A solved game is a dead ecology.
  // ------------------------------------------------------------------------
}
```

B Reproducing the numbers

All figures in §8 and §9 are exact computations over the game tree, not samples. Two scripts accompany this note. `ttt.py` enumerates positions, computes the minimax value, and defines the ladder as a priority rule list with uniform tie-breaking. `ladder.py` instruments the rule list with the price schedule of §6.1 and computes, by memoised recursion over all 255,168 plays, the joint distribution of outcome and expected metering for every pair of rungs. Symmetrised figures average a policy's results as first and as second player. The census figures are 5,478 reachable positions, 765 up to symmetry, 4,520 non-terminal, of which 1,890 offer a fork to the mover.

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